

Prop fn exp compleja igualdad fn continuas

Proposición 1. Sea $\Omega \subset \mathbb{C}$ un dominio y sean $f, g \in \mathcal{C}(\Omega)$ tales que $\exp(f) = \exp(g)$. Entonces, existe $k \in \mathbb{Z}$ tal que $f = g + 2k\pi$.

Demostración: Tenemos que $\exp(f - g) \equiv 1$, es decir,

$$\forall z \in \Omega : f(z) - g(z) \in \{2k\pi i : k \in \mathbb{Z}\}.$$

Si definimos $k(z)$ como el único entero tal que $f(z) - g(z) = 2k(z)\pi i$, tenemos que, como f y g son continuas, k es continua. Pero k es una función con valores en $\mathbb{Z}$ y Ω es conexo, luego k es constante. ■

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